

Exponential Expressions

MA.7.NSO.1.1

I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Yesterday: seven laws, one at a time.

Today: two or three laws in one expression — and the order is yours.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{10^5}{10^3}$
2. Is $\sqrt{50}$ closer to 7 or to 8?
3. Solve $4(x - 2) = 12$
4. Evaluate $2 + 3 \cdot 4^2$

Warm-Up

Answers.

1. Find the value: $\frac{10^5}{10^3}$ **100**
2. Is $\sqrt{50}$ closer to 7 or to 8? **7**
3. Solve $4(x - 2) = 12$ **$x = 5$**
4. Evaluate $2 + 3 \cdot 4^2$ **50**

Notes

Copy the table. It is your reference for the rest of the unit.

I. The seven laws, in one place

Law	Rule	Example
Product of powers	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^1 = 2^4$
Quotient of powers	$a^m \div a^n = a^{m-n}$	$(-3)^7 \div (-3)^2 = (-3)^5$
Power of a product	$(ab)^m = a^m \cdot b^m$	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$
Power of a quotient	$(a \div b)^m = a^m \div b^m$	$(13 \div 5)^3 = 13^3 \div 5^3$
Power of a power	$(a^m)^n = a^{mn}$	$(6^3)^4 = 6^{12}$
Identity exponent	$a^1 = a$	$15^1 = 15$
Zero exponent	$a^0 = 1$	$127^0 = 1$

Notes

Two students, two orders, one value.

II. More than one law — make a plan, then follow it

$$\text{Winston: } \left(\frac{2^6}{2^4}\right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left(\frac{2^6}{2^4}\right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

Notes

III. Order of operations still applies

- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:** $2^3 + 2^2 = 8 + 4 = 12$, not 2^5 .

Example 1

A zero exponent inside a product inside a quotient.

$$\frac{4^5 \cdot 4^0}{4^2}$$

Make a plan. Then find the value.

Example 1

Worked — Caroline's plan and Jamir's plan.

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2} \quad \text{Caroline: product of powers first}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2} \quad \text{Jamir: zero exponent first}$$

$$\frac{4^5}{4^2} = 4^3 = 64 \quad \text{both: quotient of powers, then evaluate}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{5^4 \cdot 5^0}{5^2}$$

Your Turn

Answer.

$$\frac{5^4 \cdot 5^0}{5^2}$$

$5^4 \cdot 1$ over 5^2 , so 5^2

Answer: $5^2 = 25$

Example 2

Three laws, a decimal and a negative base.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

Make a plan. Then find the value.

Example 2

Worked.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4 \quad \text{zero exponent; power of a quotient, backwards}$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4 \quad \text{divide inside: } -15 \div 1.5 = -10$$

$$(-10)^4 = 10000 \quad \text{four negative factors: positive}$$

Answer: 10,000

Your Turn

Same steps, your numbers. Boards up when done.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

Your Turn

Answer.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

$$1 \cdot (-12 \div 4)^3 = (-3)^3$$

Answer: -27

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{6^4 \cdot 6^0}{6^2}$$

Answer it.

Find the value.

Whiteboards • Question 1 of 9

Answer.

$$\frac{6^4 \cdot 6^0}{6^2}$$

$6^4 \cdot 1$ over 6^2 , so 6^2

Answer: 36

Whiteboards • Question 2 of 9

Boards up on three.

$$(3^0 \cdot 3^2)^3$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$(3^0 \cdot 3^2)^3$$

$$(3^2)^3 = 3^6$$

Answer: 729

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$0.5^6 \div 0.5^2 = 0.5^4$$

Answer: 0.0625

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

2.5^2 — the exponent 1 changes nothing

Answer: 6.25

Whiteboards • Question 5 of 9

Boards up on three.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. 4

B. 2^{22}

C. 0

D. $1/4$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. 4

B. 2^{22}

C. 0

D. $1/4$

Answer: A — 4

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. 0.1^5

B. $0.2^0 \cdot 0.1^2$

C. 0.1^6

D. 0.2^5

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. 0.1^5

B. $0.2^0 \cdot 0.1^2$

C. 0.1^6

D. 0.2^5

Answer: A — 0.1^5

Whiteboards • Question 7 of 9

Boards up on three.

$$0.1^2 \cdot (0.1^1)^2$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$0.1^2 \cdot (0.1^1)^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^4$$

Answer: 0.0001

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$(-4)^3$ over 1

Answer: -64

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

Write your answer in sentences.

Name the law each one used first, and give the value.

Whiteboards • Question 9 of 9

Answer.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

both plans end at 2^6

Answer: Yes — both give $2^6 = 64$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Evaluate expressions using properties of exponents — ZE9